MD. NUR ALAM



AI/ML Engineer | Computer Vision Researcher | Software Engineering | Innovator & Tech Enthusiast

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GitHub ORCID iD Md. Nur Alam Portfolio

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with a strong background in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Science, and Software Engineering. Experienced in developing machine learning and software solutions, conducting research with image-based datasets, applying transfer learning and fine-tuning, performing model evaluation, and implementing Explainable AI (XAI) techniques such as Grad-CAM. Hands on experience also covers R&D, market research, competitor intelligence, business analysis and data driven problem solving via academic and professional projects. Familiar with Python, C, Java, SQL, Git/GitHub, and current development tools, with a desire to create practical, scalable, and intelligent systems. Seeking opportunities in AI/ML Engineering, Data Science, Computer Vision, Software Engineering, and Applied AI Research.

CORE COMPETENCIES

- Artificial Intelligence & Machine Learning
- Deep Learning & Neural Networks
- Computer Vision & Image Analysis
- Transfer Learning & Model Fine-Tuning & Explainable AI (XAI)
- Data Analysis & Predictive Modeling
- Software Development & Problem Solving
- Database Design & Management
- Research & Experimental Analysis
- R&D and Technical Research
- Competitor & Market Intelligence
- Business & Data-Driven Analysis
- Team Collaboration & Technical Communication

PROFESSIONAL EXPERIENCE

Campus Bird Intern — Research & Development Department | Vacancy Hunting — Campus Bird Internship Program 9.0

July 2026 – Present | Dhaka, Bangladesh

Achievements

- Selected as a Campus Bird Intern in the Research & Development Department, placing in the Top 10% of applicants.
- Developed and presented competitor intelligence, platform analysis, and business growth strategy deliverables for R&D evaluation.
- Conducted GYC competitor analysis covering business model, customer journey, product offerings, competitive gaps, and growth opportunities.
- Conducted Skill Jobs platform research and proposed vacancy-intelligence concepts including skill-fit scoring, alerts, duplicate detection, and application tracking.

Duties & Responsibilities

- Conduct market research, competitor analysis, and platform research to identify industry trends and business opportunities.
- Collect, analyze, and interpret information to support data-driven business decisions and strategic recommendations.
- Prepare research reports, presentations, business cases, and actionable insights for R&D activities.
- Identify opportunities for AI integration, workflow automation, predictive analytics, and process improvement.

Selected Apprentice Programmer | Daffodil Family Apprenticeship Program 2026 — HRDI, Daffodil International University

July 2026 – Present | Dhaka, Bangladesh

Achievements

- Selected through a competitive multi-stage selection process for the Daffodil Family Apprenticeship Program 2026.
- Secured Top 8 in the 100-mark assessment and ranked 18th among 38 final selected candidates.
- Completed a 3-day Apprenticeship Bootcamp focused on problem-solving, business case analysis, teamwork, and professional presentation.
- Completed Building Skills for TAF & Apprenticeship with Distinction from GoEdu.

Duties & Responsibilities

- Performed business case analysis using root-cause analysis, SWOT analysis, KPI evaluation, and process analysis.
- Identified opportunities for AI integration, workflow automation, predictive analytics, and data-driven decision-making.
- Participated in team discussions, peer reviews, business case activities, and Shark Tank-style presentations.
- Applied structured business observation to identify operational challenges, process improvements, and automation opportunities.

EDUCATIONAL QUALIFICATIONS

Bachelor of Science in Computer Science and Engineering
Daffodil International University, Dhaka | **CGPA:** 3.63/4.00 | Apr 2022 – May 2026

Higher Secondary Certificate (HSC) – Science
Shahid A.H.M. Kamaruzzaman Govt. Degree College, Rajshahi | **GPA:** 5.00/5.00 | Year: 2020

Secondary School Certificate (SSC) – Science
Al Helal Islami Academy & College, Naogaon | **GPA:** 5.00/5.00 |Year: 2017

PROJECTS & ACADEMIC ACTIVITIES

1. Amrapali Mango Fruit Disease Classification & Prediction using Transfer Learning with (XAI)

Python, TensorFlow, CNN, Transfer Learning, Grad-CAM, Computer Vision

- Developed a cultivar-specific deep learning system for classifying 7 mango fruit diseases using transfer learning and fine-tuning.
- Applied image preprocessing, augmentation, model evaluation, and Explainable AI (Grad-CAM) for visual interpretation.

2. Customer Relationship Management System

Python, Streamlit, SQLAlchemy, SQLite/MySQL, OOP

- Built a CRM application for customer management, lead/sales pipeline, follow-ups, support tickets, and analytics.
- Applied Object-Oriented Programming, database management, authentication, and audit logging.

3. Hospital Management System

PHP, MySQL, HTML, CSS, JavaScript, Bootstrap, AJAX

- Developed a multi-user web application for hospital administration, doctor/patient management, appointments, and medical records.
- Implemented database-driven workflows and role-based functionality using PHP and MySQL.

4. Hotel Management System

Python, MySQL, Streamlit, FastAPI, SQLAlchemy

- Developed a role-based hotel management system covering reservations, check-in/out, payments, invoices, reports, and audit logging.
- Implemented database-driven workflows, REST APIs, authentication, and business logic.

RESEARCH & PUBLICATIONS

Undergraduate Thesis
Explainable Transfer Learning-Based Deep Convolutional Neural Network for Cultivar-Specific Amrapali Mango Fruit Disease Classification — Daffodil International University | 2025–2026

Research Paper — Accepted for Presentation
Benchmarking Attention-Enhanced Encoder–Decoder Models for Polyp Segmentation in Colonoscopy Images
IEEE RAAICON 2026 | Paper ID: 433

Research Dataset — Mendeley Data
Amrapali Mango Fruit Diseases: A Cultivar-Specific Image Dataset for Computer Vision and Deep CNN Classification
Mendeley Data | 2 January 2026 | **DOI: 10.17632/ypttkp5fb5.1**

LEADERSHIP ACTIVITIES

- Department Representative — ResearchOrbit | 2025–2026
- Media Secretary — DIU CPC | 2024–2025
- Executive Member, Research Wing — DIU CPC | 2023–2024
- Associate Member, Research Wing — DIU CPC | 2022–2023
- Campus Leader — SelDev Academy | 2023–2024
- Logistics & Protocol Volunteer — DIU Job Utsob 2025 | Nov 28–29, 2025
- Official Photographer & Volunteer — ICPC Asia Dhaka Regional Contest, 2024
- Photographer & Documentary Volunteer — National Undergraduate Mathematical Olympiad 2025
- Photographer — JCPC 2024

AWARDS & CERTIFICATIONS

- Building Skills for TAF & Apprenticeship — With Distinction, GoEdu, 2026
- Certificate of Appreciation — Erasmus Mundus Joint Masters & Scholarship Webinar, DIU Research Society, 2025
- Media Secretary — DIU Computer & Programming Club, 2024–2025
- Logistics & Protocol Volunteer — DIU Job Utsob 2025
- Volunteer — ICPC Asia Dhaka Regional Contest 2024
- Participant — Unlock the Algorithm Programming Contest (Spring 2024)
- Participant — Engineering Summit, IDEB Bhaban (YBFS), 2023
- Participant — Take-Off Programming Contest (Fall 2022)
- Academic Merit Scholarship, DIU—Up to 50% Tuition Waiver (2022-2026)
- JSC Merit Scholarship, Rajshahi Board (2014)

TECHNICAL SKILLS

- Programming: Python, C, Java, PHP, JavaScript
- AI/ML: Machine Learning, Deep Learning, Computer Vision, CNN, Transfer Learning, XAI, Grad-CAM
- Data Science: NumPy, Pandas, Matplotlib, Scikit-learn
- Development: OOP, DSA, REST API, Streamlit, FastAPI
- Database: SQL, MySQL
- Tools: Git, GitHub, Docker, VS Code, PyCharm
- Systems & Networking: Linux/Unix, Cisco Packet Tracer
- Engineering: Software Development, Debugging, LaTeX, Agile/Scrum

REFERENCES

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